

Dynamic Fourier Saddles Beyond the Kernel Regime: Full-Training Depth, Low Rank, and Spectral-Power Descent

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Abstract

We study population training on the circle for finite-depth fully connected networks and low-rank matrix-factorized neural networks (MMNNs), in both a right-factor-only and a full-training maximum-update scaling. In the masked model an exact finite-width identity factorizes the instantaneous tangent kernel as $K_t(x, x') = \Phi_0(x, x')G_t(x, x')$: the first-feature kernel is fixed while a back-propagated gate kernel evolves. In full training, every activation and backward field evolves and must instead be closed by deep dynamical mean-field theory (DMFT).

Our main theorem is architecture- and depth-general and does not freeze the kernel. At every point of a nonlinear trajectory, bounded-variation tangent features imply $\Lambda_{qq}(t) \leq \mathcal{B}_t/q^2$. Hence a residual concentrated on $\cos(qx)$ is an approximate saddle with inverse-gradient index at least $q^2/\mathcal{B}_t$; a quantitative leakage condition gives a quadratic residence-time lower bound. This is a spectral regularity obstruction, not the coordinate-support leap complexity of Abbe et al. [1]: after the single input coordinate is exposed, their leap cannot distinguish frequency four from frequency sixteen. Layerwise, the exact tangent feature is a forward activation times a backward field, so a mode becomes accessible only when their evolving Fourier spectra acquire an aligned convolution channel.

We then analyze exact spectral-power descent $G = U\Sigma V^\top \mapsto U\Sigma^p V^\top$, including polar Muon ($p = 0$) and Muon^p. The update is always a strict descent direction. Under an explicit orthogonal-sector closure, it changes the half-recovery exponent from $2s$ for gradient flow to $s(p + 1) - \alpha(1 - p)$ for a target spectrum $a_q \asymp q^{-\alpha}$ and tangent amplitudes q^{-s} . This gives a mechanism by which spectral flattening advances a weak Fourier front. The corresponding reused-data Muon DMFT requires a nonseparable matrix response and Onsager memory and is stated conditionally, not presented as an existing theorem.

Paired full-batch experiments train every block of affine-depth 3, 5, 7 dense networks and signed low-rank MMNNs. They confirm ordered Fourier-sector recovery with order-one feature motion, while exact Muon and fractional powers can move high-frequency clocks earlier on both sparse cosine sums and power-law spectra. The gain is optimizer-, architecture-, and seed-dependent and is not a universal terminal-risk dominance. Code, raw trajectories, calibration records, and vector figures are included.

1 Introduction

Neural networks often fit low-frequency components before high-frequency components. In a fixed neural tangent kernel (NTK), the explanation is linear: Fourier modes with smaller kernel eigenvalues decay more slowly [4, 11]. That explanation is insufficient under maximum-update parameterization. Standard and NTK parameterizations do not retain feature learning at infinite width, whereas μP does [12]. In the rich limit the kernel evolves with the features, and its evolution must be determined self-consistently [5].

This raises a precise question. Suppose the target is

$$y(x) = \sum_{j=1}^J a_j \cos(q_j x), \quad 1 \leq q_1 < \dots < q_J, \quad (1)$$

and training displays long plateaux between the recovery of successive modes. Are these plateaux merely an NTK artifact, or can one establish a hierarchy of frequency-indexed saddles in a genuinely feature-learning limit?

The word “leap” requires care. The leap complexity of Abbe et al. [1] is a combinatorial measure of which new coordinates of an isotropic high-dimensional target become detectable. It predicts sequential support learning and saddle-to-saddle dynamics. It is not the integer difference $q_j - q_{j-1}$ between two Fourier frequencies. The one-dimensional periodic problem studied here has an analogous sequential phenomenology but a different obstruction: Fourier scarcity of the tangent features. A major conclusion of this paper is that a universal law in the raw difference $q_j - q_{j-1}$ is false. The geometric obstruction is indexed by the absolute frequency q_j .

We begin with the low-rank multi-component neural network (MMNN)

$$W_2(t) = \frac{1}{\sqrt{mr}} UV(t)^\top, \quad (2)$$

where $U \in \mathbb{R}^{n \times r}$ is frozen and only the right factor $V \in \mathbb{R}^{m \times r}$ is trained. The rank is extensive but low: $r/m \rightarrow \rho \in (0, 1)$. This distinction is essential. If r is fixed, the finite collection of rank channels does not self-average and there is no deterministic DMFT without retaining channel randomness.

We then remove the mask. Every dense matrix, both factors of every MMNN matrix, every bias, and the readout are trained at affine depths three, five, and seven. This distinction matters twice. It tests whether the saddle hierarchy is an artifact of the frozen-left architecture, and it exposes the optimizer question: matrix spectral descent can redistribute motion across the singular sectors of each full gradient in a way that scalar-metric gradient flow cannot.

Our contributions are as follows.

1. We give an exact finite-width factorization $K_t = \Phi_0 G_t$ for right-factor training. It isolates the DMFT order parameter that can change: the back-propagated gate kernel G_t .
2. We prove an instantaneous q^{-2} upper bound on the tangent curvature at *every* point of any finite-depth nonlinear trajectory whose tangent features have bounded variation. This yields a hierarchy of approximate saddles for both masked and fully trained networks without freezing K_t . An exact forward–backward product identity shows that the missing Fourier supply is a convolution of the two learned spectra, projected through rank channels in an MMNN.
3. We give a conditional residence-time theorem that remains valid for a time-dependent, non-diagonal kernel. Its assumptions quantify precisely the Fourier leakage that must be checked rather than silently assuming translation invariance after task-dependent training.
4. We prove the descent identity for a normalized singular-value power family and, under a declared orthogonal-sector closure, derive its Fourier front exponent for power-law targets. We identify the matrix response that a Muon DMFT must close and separate it from established gradient-flow DMFT.
5. We test width, depth, rank, absolute frequency, raw gap, full versus masked training, and exact Muon powers in paired full-batch experiments. The masked model supplies a counterexample in which rich learning deepens high-frequency saddles; full training and spectral flattening can reverse the clock ordering without guaranteeing a better endpoint.

Scope. We use “matrix factorized” rather than literal nonnegative matrix factorization. The factors are signed. A strict nonnegative constraint changes gradient flow into a tangent-cone or projected flow and is outside the theorems. We also use “saddle” in the quantitative approximate sense of a state with small gradient relative to a nonzero residual; generically these plateaux are not exact critical points.

2 Architecture and masked $\mu\mathbf{P}$ scaling

Let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$, equipped with uniform probability measure. We use the normalized inner product

$$\langle u, v \rangle = \frac{1}{2\pi} \int_0^{2\pi} u(x)v(x)dx.$$

The first features are

$$h_j(x) = \sqrt{2} \operatorname{ReLU}(\cos(x - \omega_j) + b_j), \quad j \leq m, \quad (3)$$

with i.i.d. random phases and biases. Define the rank channels, outer preactivations, and centered output by

$$z_a(x, t) = \frac{1}{\sqrt{m}} \sum_{j=1}^m V_{ja}(t) h_j(x), \quad (4)$$

$$s_i(x, t) = \frac{1}{\sqrt{r}} \sum_{a=1}^r U_{ia} z_a(x, t) + \beta_i, \quad (5)$$

$$f_t(x) = \frac{1}{\gamma n} \sum_{i=1}^n c_i [\phi(s_i(x, t)) - \phi(s_i(x, 0))]. \quad (6)$$

Here $U_{ia}, V_{ja}(0) \sim N(0, 1)$, the readouts $c_i \in \{-1, 1\}$ are symmetric, and $\phi = \operatorname{ReLU}$. Centering makes $f_0 = 0$ exactly and removes the finite-width output fluctuation that otherwise grows as $\gamma \downarrow 0$. It does not alter any derivative with respect to V .

For square population loss

$$\mathcal{L}(V) = \frac{1}{2} \|f_V - y\|_{L^2(\mathbb{T})}^2, \quad R_t = f_t - y, \quad (7)$$

we train only V using the metric gradient flow

$$\dot{V}(t) = -\gamma^2 n \nabla_V \mathcal{L}(V(t)). \quad (8)$$

With m, n, r proportional, $\gamma = \Theta(1)$ produces an order-one change in the rank-channel fields. This is the maximal-update scaling for the trainable matrix, with all other layerwise learning-rate multipliers set to zero. We call it *masked $\mu\mathbf{P}$* . It should not be confused with full $\mu\mathbf{P}$, in which every layer is trained maximally.

The scalar γ is the feature-learning strength used in the DMFT parameterization of Bordelon and Pehlevan [5]. Taking the wide limit at fixed $\gamma > 0$ gives feature learning; taking $\gamma \downarrow 0$ afterward gives the lazy limit. This order of limits is important.

2.1 Full-training dense and factorized networks

The full-training controls use the periodic input embedding $u(x) = (\cos x, \sin x)$. Let $D \geq 3$ denote the number of affine maps, including the scalar readout. A dense network is

$$h^1(x) = \text{ReLU}\left(Au(x)/\sqrt{2} + b^1\right), \quad (9)$$

$$h^{\ell+1}(x) = \text{ReLU}\left(W^\ell h^\ell(x)/\sqrt{m} + b^{\ell+1}\right), \quad 1 \leq \ell \leq D-2, \quad (10)$$

$$\tilde{f}_\theta(x) = m^{-1}c^\top h^{D-1}(x) + a_0, \quad f_\theta(x) = \tilde{f}_\theta(x) - \tilde{f}_{\theta_0}(x). \quad (11)$$

All entries of the raw matrices and readout are initialized as standard Gaussians. The explicit factors $m^{-1/2}$ in the hidden maps and m^{-1} in the readout carry the width scaling.

For a signed low-rank MMNN, every square hidden map is replaced by

$$\frac{1}{\sqrt{m}}W^\ell h = \frac{1}{\sqrt{mr}}U^\ell(V^\ell)^\top h, \quad U^\ell, V^\ell \in \mathbb{R}^{m \times r}, \quad r/m = \rho. \quad (12)$$

Both factors are trainable. The input map, biases, and readout are also trainable, so this is not the mask of Equation (8). The factorization is again signed rather than nonnegative.

Write θ_ν for a scalar raw parameter. The full-training metric is

$$\dot{\theta}_\nu = -\eta_\nu \partial_{\theta_\nu} \mathcal{L}, \quad \eta_\nu = \begin{cases} m, & \theta_\nu \neq a_0, \\ 1, & \theta_\nu = a_0. \end{cases} \quad (13)$$

This is a convenient raw-coordinate form of maximal update. For example, a hidden raw-matrix gradient is order $m^{-3/2}$; multiplying it by m gives an order $m^{-1/2}$ entrywise change whose coherent forward action is order one. The input weights, hidden biases, factors, and readout obey the analogous balance. Width-transfer experiments check rather than assume that the resulting feature trajectories have a common time scale.

3 Exact dynamics and the DMFT reduction

The prediction dynamics always admit a tangent-kernel representation. This identity alone does not make training a kernel method: a kernel method uses a kernel fixed independently of the trajectory, whereas the kernel below is a state variable.

Define

$$\Phi_m(x, x') = \frac{1}{m} \sum_{j=1}^m h_j(x) h_j(x'), \quad (14)$$

$$g_a(x, t) = \frac{1}{\sqrt{n}} \sum_{i=1}^n c_i U_{ia} \phi'(s_i(x, t)), \quad (15)$$

$$G_{r,t}(x, x') = \frac{1}{r} \sum_{a=1}^r g_a(x, t) g_a(x', t). \quad (16)$$

Theorem 3.1 (Exact right-factor kernel factorization). *At every finite width and every differentiability time of the ReLU flow,*

$$K_t(x, x') := \gamma^2 n \langle \nabla_V f_t(x), \nabla_V f_t(x') \rangle_F = \Phi_m(x, x') G_{r,t}(x, x'), \quad (17)$$

$$\partial_t f_t(x) = -\langle K_t(x, \cdot), R_t \rangle, \quad \dot{\mathcal{L}}(t) = -\langle R_t, K_t R_t \rangle, \quad (18)$$

where $(\mathcal{K}_t u)(x) = \langle K_t(x, \cdot), u \rangle$. Moreover,

$$\partial_t z_a(x, t) = -\gamma \sqrt{\frac{n}{r}} \langle R_t(x') \Phi_m(x, x'), g_a(x', t) \rangle_{x'}. \quad (19)$$

Proof. The chain rule gives

$$\frac{\partial f_t(x)}{\partial V_{ja}} = \frac{h_j(x) g_a(x, t)}{\gamma \sqrt{nmr}}.$$

Multiplying the Frobenius inner product of two such gradients by $\gamma^2 n$ and summing over (j, a) yields (17). Equations (18) follow from metric gradient flow. Finally, insert the same derivative into (8), multiply by $h_j(x)/\sqrt{m}$, and sum over j . $\square$

Equation (17) identifies the source of non-kernel frequency learning. The phase-averaged limit of Φ_m is a stationary first-feature kernel and is frozen. All task-dependent adaptation enters through $G_{r,t}$, which records the collective outer-gate and frozen-left factor response. In a fully trained deep network, DMFT uses two-time activation and gradient kernels for every layer; here the mask removes all terms except the order parameter paired with the trained right factor.

3.1 The extensive-rank DMFT limit

Let

$$\frac{n}{m} \rightarrow \alpha \in (0, \infty), \quad \frac{r}{m} \rightarrow \rho \in (0, 1], \quad \frac{n}{r} \rightarrow \kappa = \frac{\alpha}{\rho}. \quad (20)$$

For a finite training set and a smooth activation with polynomially bounded derivatives, tensor-program limit theorems imply concentration of empirical activation and gradient inner products at every finite collection of times [12]. The equivalent DMFT description is a self-consistent stochastic process with two-time correlations and response functions [5]. Applying that result to the computation graph (4)–(8), with zero learning-rate masks on the frozen layers, gives the following specialization.

Proposition 3.2 (Masked DMFT; specialization of existing wide-limit theorems). *Under the preceding hypotheses and for fixed time horizon, the empirical order parameters Φ_m , $G_{r,t}$, the channel covariance $Q_t(x, x') = r^{-1} \sum_a z_a(x, t) z_a(x', t)$, and their two-time analogues converge in probability to deterministic limits. Their self-consistent single-channel process is generated by the integrated form of (19), together with the Gaussian frozen matrices and the response of (15) to z . The limiting prediction obeys*

$$\partial_t f_t = -\mathcal{K}_t(f_t - y), \quad K_t = \Phi_0 G_t, \quad (21)$$

where G_t , rather than G_0 , is solved self-consistently. Finite-width fluctuation theory for the corresponding mean-field order parameters predicts inverse-square-root corrections away from critical amplification [6].

The proposition imports the wide-limit result; it is not a new proof of the general tensor-program master theorem. For ReLU, one may either use its standard smooth approximation and pass to the limit away from zero preactivations, or formulate convergence at almost every time. The finite-width identities and the Fourier theorem below require neither Gaussianity nor a wide limit.

Equation (19) also makes the parameterization distinction transparent. For bounded order parameters on a fixed time interval, $z(t) - z(0) = O(\gamma)$; hence $G_t - G_0 \rightarrow 0$ as $\gamma \rightarrow 0$. At fixed $\gamma = \Theta(1)$, the prefactor $\gamma \sqrt{n/r} = \gamma \sqrt{\kappa}$ is order one. Narrower extensive bottlenecks therefore strengthen the field response, but this does not determine its sign in any Fourier direction.

Remark 3.3 (Why fixed rank is different). If $r = O(1)$, $G_{r,t}$ remains an average of finitely many random channel processes. It need not converge to a deterministic kernel even if $m, n \rightarrow \infty$. Fixed-rank experiments are meaningful, but their theory is a random finite-channel mean-field process, not [Proposition 3.2](#).

4 A non-kernel hierarchy of Fourier saddles

Let $e_q(x) = \sqrt{2} \cos(qx)$, so that $\|e_q\| = 1$. Let $\theta = (\theta_\nu)_{\nu \leq P}$ collect any finite set of trained scalar parameters and let $M = \text{diag}(\eta_1, \dots, \eta_P) \succ 0$ be its gradient metric. Thus $M = \gamma^2 n I$ for the masked model and M is given by [Equation \(13\)](#) for full training. We write $\|v\|_M^2 = v^\top M v$. Define

$$K_t(x, x') = \nabla_\theta f_t(x)^\top M \nabla_\theta f_t(x'), \quad \chi_{\nu,t}(x) = \sqrt{\eta_\nu} \partial_{\theta_\nu} f_t(x), \quad (22)$$

$$\Lambda_{q\ell}(t) = \langle e_q, \mathcal{K}_t e_\ell \rangle = \nabla_\theta \langle f_t, e_q \rangle^\top M \nabla_\theta \langle f_t, e_\ell \rangle. \quad (23)$$

For the masked architecture, the generic tangent feature reduces exactly to

$$\chi_{ja,t}(x) = \gamma \sqrt{n} \frac{\partial f_t(x)}{\partial V_{ja}} = \frac{h_j(x) g_a(x, t)}{\sqrt{mr}}. \quad (24)$$

This matrix need not be diagonal. Although the initialization law is translation invariant, the target (1) breaks translation symmetry during feature learning. Treating the Fourier modes as exact eigenfunctions of K_t would therefore smuggle a kernel-regime assumption into the argument.

For off-diagonal statements below, the index ℓ ranges over the complete normalized real Fourier basis $\{1, \sqrt{2} \cos(kx), \sqrt{2} \sin(kx) : k \geq 1\}$; the symbol q is shorthand for its cosine element $\sqrt{2} \cos(qx)$. Thus the leakage term includes constant and sine components created by a finite random network.

Assumption 4.1 (Finite tangent variation at a plateau). For almost every time under consideration, each periodic function $\chi_{\nu,t}$ is of bounded variation and

$$\mathcal{B}_t := \frac{1}{2\pi^2} \sum_{\nu=1}^P \text{TV}(\chi_{\nu,t})^2 < \infty. \quad (25)$$

For a finite ReLU network this holds generically whenever the outer preactivations have finitely many transversal zero crossings. The value of $\mathcal{B}_t$ is allowed to change by order one; that freedom is precisely what keeps the result outside the NTK regime.

Theorem 4.2 (Instantaneous dynamic Fourier-saddle bound). *Under [Assumption 4.1](#), at every differentiability time and for every $q \geq 1$,*

$$0 \leq \Lambda_{qq}(t) \leq \frac{\mathcal{B}_t}{q^2}. \quad (26)$$

If $R_t = A e_q$, then

$$\|\nabla_\theta \mathcal{L}(\theta_t)\|_M^2 = A^2 \Lambda_{qq}(t) \leq \frac{A^2 \mathcal{B}_t}{q^2}. \quad (27)$$

Consequently the inverse-gradient saddle index

$$S_q(t) := \frac{\|R_t\|^2}{\|\nabla_\theta \mathcal{L}(\theta_t)\|_M^2} \quad (28)$$

satisfies $S_q(t) \geq q^2 / \mathcal{B}_t$.

Proof. For a periodic bounded-variation function u , integration by parts in the Riemann–Stieltjes sense gives

$$|\langle u, e_q \rangle| = \left| \frac{\sqrt{2}}{2\pi q} \int_0^{2\pi} \sin(qx) dD u(x) \right| \leq \frac{\sqrt{2}}{2\pi q} \text{TV}(u).$$

By (22), $\Lambda_{qq} = \sum_\nu |\langle \chi_{\nu,t}, e_q \rangle|^2$. Squaring the last display and summing proves (26). If $R_t = Ae_q$, metric gradient flow and (18) give $\|\nabla_\theta \mathcal{L}\|_M^2 = A^2 \Lambda_{qq}$. The saddle-index bound follows. $\square$

Corollary 4.3 (Finite-depth full-training networks). *At every generic differentiability time, Theorem 4.2 applies to the fully trained dense and factorized networks (9)–(12) at every finite affine depth and finite width.*

Proof. On each activation cell, every forward field is a finite linear combination of restrictions of trigonometric functions, and every backward field is a finite product of such fields with ReLU gate indicators. At a generic parameter value no preactivation vanishes identically on a nontrivial interval and all zero crossings are finite and transversal. Forward fields, gate indicators, their finite products, and therefore all tangent features have bounded variation. Signed factorization changes the finite linear combinations but not this closure property. $\square$

The theorem is deliberately instantaneous. It permits K_t , its Fourier matrix, the gates, and $\mathcal{B}_t$ all to evolve. A residence-time claim additionally needs control of off-diagonal transport.

Remark 4.4 (Wide-limit qualification). Finite-width bounded variation does not by itself imply $\sup_m \mathcal{B}_t^{(m)} < \infty$ for a sequence of widening networks. If the maximal-update sequence obeys this uniform regularity bound on a fixed time interval, then (26) passes immediately to every limiting DMFT curvature. Establishing that bound from the full two-time DMFT is a separate problem: generic finite-network BV is not a substitute for it. Our main-width diagnostics measure Λ_{qq} directly and retain high-frequency decay, while width transfer supports a shared training scale; neither observation proves uniform control of the larger variation budget.

Assumption 4.5 (Plateau purity and Fourier leakage). On a time interval $I = [t_0, t_1]$, a next mode q satisfies

$$\left| \sum_{\ell \neq q} \Lambda_{q\ell}(t) R_\ell(t) \right| \leq \varepsilon \Lambda_{qq}(t) |R_q(t)|, \quad \sup_{t \in I} \mathcal{B}_t \leq \mathcal{B}_*, \quad (29)$$

where $R_\ell = \langle R, e_\ell \rangle$ and $0 \leq \varepsilon < 1$.

Theorem 4.6 (Residence time with a dynamic, non-diagonal kernel). *Under Assumption 4.5, if $|R_q|$ decreases from A to $B < A$ while the trajectory remains in I , then*

$$t_1 - t_0 \geq \frac{q^2}{(1 + \varepsilon) \mathcal{B}_*} \log \frac{A}{B}. \quad (30)$$

Proof. Project $\partial_t R = -\mathcal{K}_t R$ on e_q . By (29) and Theorem 4.2,

$$|\partial_t R_q| \leq (1 + \varepsilon) \Lambda_{qq} |R_q| \leq (1 + \varepsilon) \frac{\mathcal{B}_*}{q^2} |R_q|.$$

Integrating the logarithmic derivative proves the claim. $\square$

Corollary 4.7 (Dynamic saddle bound in a feature-learning DMFT limit). *Consider a sequence of widths for which the maximal-update prediction and tangent curvatures converge on a fixed time interval to the Bordelon–Pehlevan DMFT solution. If $\sup_{m,t} \mathcal{B}_t^{(m)} \leq \bar{\mathcal{B}} < \infty$, then the limiting, generally time-dependent curvature satisfies*

$$\Lambda_{qq}^{\text{DMFT}}(t) \leq \frac{\bar{\mathcal{B}}}{q^2}. \quad (31)$$

If the limiting Fourier leakage satisfies [Assumption 4.5](#), with $\mathcal{B}_$ replaced by $\bar{\mathcal{B}}$, then its residence-time bound is [\(30\)](#).*

Proof. Apply [Theorem 4.2](#) at each finite width and pass to the curvature limit. The residence statement follows by the projected residual argument of [Theorem 4.6](#). $\square$

Corollary 4.8 (Hierarchy and the meaning of a frequency leap). *Suppose the modes in [\(1\)](#) become successively residual-dominant, and [Assumption 4.5](#) holds with common $\varepsilon, \mathcal{B}_*$ on their plateaux. Then the lower bound on the j th residence time is proportional to q_j^2 . If q_{j-1} is held fixed, a larger raw gap implies a larger bound. Without fixing q_{j-1} , no law depending only on $q_j - q_{j-1}$ follows.*

This is the rigorous content of “the bigger the frequency leap, the bigger the saddle.” The statement is correct when “leap” means the absolute new frequency, or when the predecessor is fixed. A translation of both frequencies by the same integer preserves their gap but changes the bound.

4.1 Why coordinate leap complexity does not supply this hierarchy

The similar saddle-to-saddle phenomenology should not obscure a mismatch of complexity measures. The formal setting of Abbe et al. [\[1\]](#) is a tensor-product Hermite or Fourier–Walsh basis on isotropic high-dimensional data, not the continuous Fourier basis on $\mathbb{T}$. Even in the closest one-coordinate Hermite analogue, the higher degree is not the next leap.

Proposition 4.9 (One-coordinate leap collapse). *Let $h(z) = \sum_{j=1}^J a_j \text{He}_{q_j}(z_1)$ with nonzero coefficients and distinct positive degrees. Under [Definition 1](#) of Abbe et al. [\[1\]](#),*

$$\text{Leap}(h) = \min_j q_j. \quad (32)$$

In particular, $\text{He}_1 + \text{He}_4 + \text{He}_{16}$ has leap one; the framework assigns no new coordinate leap to degrees four and sixteen.

Proof. Whichever basis element appears first in an admissible ordering introduces the only active coordinate, counted with its Hermite multiplicity. Once coordinate one belongs to the accumulated support, every later basis element introduces zero unused coordinates under the set-difference rule. The maximum increment for an ordering is therefore the degree of its first element, and minimizing over orderings gives [\(32\)](#). $\square$

Our Fourier saddle bound is thus a *within-coordinate regularity cost*. For a genuinely high-dimensional periodic target, coordinate support discovery and within-support oscillation can coexist. A plausible composite complexity would retain the coordinate leap for discovering a new support and attach a Fourier-regularity budget to the new sector. Neither factor is a theorem-level substitute for the other.

5 What DMFT adds beyond the theorem

Theorems 4.2 and 4.6 identify a geometric obstruction but do not prescribe the trajectory of $\mathcal{B}_t$ or $\Lambda(t)$. DMFT supplies that missing dynamics through G_t . Combining (17) with (23) gives

$$\Lambda_{q\ell}(t) = \int_{\mathbb{T}^2} e_q(x) \Phi_0(x, x') G_t(x, x') e_\ell(x') d\mu(x) d\mu(x'). \quad (33)$$

There are two non-kernel escape channels.

1. *Diagonal amplification*: gate motion increases $\Lambda_{qq}(t)$, spending more tangent-variation budget at the missing frequency.
2. *Off-diagonal transfer*: task adaptation creates $\Lambda_{q\ell}(t) R_\ell(t)$ large enough to violate Assumption 4.5 and feed mode q from other residual components.

Neither channel has a fixed sign. The same gate rearrangement can align features with a low-frequency component while removing high-frequency zero crossings, thereby reducing Λ_{qq} for large q . The right-factor-only mask is especially restrictive: V can move the gates, but the first dictionary, left factor, and readout cannot rotate to repair a harmful redistribution. This predicts the suppression observed in Section 7.

For full training, the corresponding statement is layerwise. Suppressing normalization constants, introduce the two-time forward and backward order parameters

$$\begin{aligned} C_{t,s}^\ell(x, x') &= \lim_{m \rightarrow \infty} \frac{1}{m} \left\langle h_t^\ell(x), h_s^\ell(x') \right\rangle_{\mathbb{R}^m}, \\ D_{t,s}^\ell(x, x') &= \lim_{m \rightarrow \infty} \frac{1}{m} \left\langle b_t^\ell(x), b_s^\ell(x') \right\rangle_{\mathbb{R}^m}. \end{aligned} \quad (34)$$

where b_t^ℓ is the width-normalized back-propagated field, together with the causal responses of each field to its effective Gaussian drive. At equal times the dense tangent kernel has the schematic layer decomposition

$$K_t(x, x') = K_t^{\text{readout}+\text{bias}}(x, x') + \sum_{\ell} a_\ell C_{t,t}^{\ell-1}(x, x') D_{t,t}^\ell(x, x'), \quad (35)$$

where the positive constants a_ℓ are fixed by the maximal-update metric. The factorized network augments this finite list with left- and right-channel covariances and responses. Bordelon–Pehlevan DMFT closes these objects as coupled causal Volterra processes. Consequently

$$\dot{r}_q(t) = - \sum_{\ell'} \Lambda_{q\ell'}[C, D, \mathcal{R}](t) r_{\ell'}(t) \quad (36)$$

is nonlinear through the order parameters even though it is linear in the instantaneous residual. A frequency plateau is the interval during which this self-consistent process has not supplied diagonal curvature or off-diagonal transfer to the missing sector. This is the DMFT content of the feature-learning explanation; replacing $[C, D, \mathcal{R}](t)$ by its initial value would reduce it to NTK dynamics.

5.1 The exact source of Fourier supply

The preceding “supply” has a concrete layerwise meaning. Use the complex basis $\varphi_q(x) = e^{iqx}$, and let $b_i^{\ell+1}(x, t)$ denote the derivative of the output with respect to the preactivation of unit i after the hidden matrix W^ℓ .

Proposition 5.1 (Exact forward–backward Fourier supply). *In the raw maximal-update coordinates of Equation (13), the normalized tangent feature of a dense hidden weight is*

$$\chi_{ij,t}^{W^\ell}(x) = \sqrt{m} \partial_{W_{ij}^\ell} f_t(x) = b_i^{\ell+1}(x, t) h_j^\ell(x, t). \quad (37)$$

Consequently its Fourier coefficient is the convolution

$$\widehat{\chi}_{ij,t}^{W^\ell}(q) = \sum_{k \in \mathbb{Z}} \widehat{b}_i^{\ell+1}(k, t) \widehat{h}_j^\ell(q - k, t), \quad \Lambda_{qq}^{W^\ell}(t) = \sum_{i,j} \left| \widehat{\chi}_{ij,t}^{W^\ell}(q) \right|^2. \quad (38)$$

The analogous identities for a factorized map are

$$\begin{aligned} \chi_{ia,t}^{U^\ell}(x) &= r^{-1/2} b_i^{\ell+1}(x, t) [(V^\ell)^\top h^\ell(x, t)]_a, \\ \chi_{ja,t}^{V^\ell}(x) &= r^{-1/2} h_j^\ell(x, t) [(U^\ell)^\top b^{\ell+1}(x, t)]_a. \end{aligned} \quad (39)$$

Proof. Differentiating $W^\ell h^\ell / \sqrt{m}$ with respect to W_{ij}^ℓ , and multiplying by the metric factor $\sqrt{m}$, gives (37). The Fourier product formula gives the first identity in (38); summing its squared coefficient over the block gives its tangent-curvature contribution. Differentiating $U^\ell (V^\ell)^\top h^\ell / \sqrt{mr}$ with respect to each factor and again multiplying by $\sqrt{m}$ gives (39). $\square$

These are finite-width identities, not a Gaussian or kernel approximation. They show that a high mode q is weak when the current forward and backward spectra contain few well-aligned pairs whose frequencies sum to q . Learning lower modes changes both spectra and can therefore create a new convolution channel, rotate a low-rank channel into alignment, or destroy one. Writing

$$\dot{r}_q = - \underbrace{\Lambda_{qq} r_q}_{\text{diagonal supply}} - \underbrace{\sum_{\ell \neq q} \Lambda_{q\ell} r_\ell}_{\text{transport supply}} \quad (40)$$

makes the plateau criterion explicit: both supplies are small compared with the unresolved residual. A feature-learning transition is the self-consistent growth of either term through (38)–(39). The q^{-2} theorem limits the diagonal route under uniform regularity; the leakage assumption limits the transport route. This is why the hierarchy can persist while the tangent kernel changes by order one.

Deep linear DMFT gives analytic examples in which feature learning accelerates power-law training [7]. Data-dependent fixed-point kernels can likewise outperform their lazy counterparts when all relevant feature degrees of freedom adapt [10]. Those results do not imply monotone acceleration for a masked nonlinear architecture. Our experiment is a counterexample to that extrapolation.

The equilibrium analysis of Barbier et al. [3] is complementary. It uses replicas and Harish–Chandra–Itzykson–Zuber (HCIZ) integrals to describe Bayes-optimal specialization transitions in proportional-width teacher–student networks, and it predicts metastable unspecialized solutions. HCIZ is not required for (33): we condition on the frozen U , study gradient-flow dynamics rather than an equilibrium posterior, and never integrate a learned rotational degree of freedom. That work motivates the language of specialization and metastability, but using its static free energy as a training-time potential here would be unjustified.

6 Spectral-power descent and the Muon front

Muon replaces the singular values of a matrix gradient while preserving its singular frames [9]. To isolate this operation from momentum, weight decay, and an approximate Newton–Schulz polynomial, we use the exact memoryless map. For a matrix block $B \in \mathbb{R}^{d_o \times d_i}$ with gradient $G_B = U\Sigma V^\top$, define

$$\Psi_p(G_B) = U\Sigma^p V^\top, \quad \Phi_p(G_B) = \sqrt{d_o d_i} \frac{\Psi_p(G_B)}{\|\Psi_p(G_B)\|_F}, \quad 0 \leq p \leq 1. \quad (41)$$

Zero singular values remain zero. The member $p = 0$ is the compact polar factor; fractional members are Muon^p [8]. The RMS normalization makes a width-independent learning rate plausible in a maximal-update scaling; it is not, by itself, a proof of hyperparameter transfer [13]. Vectors and scalar biases retain the metric gradient update. Up to scale, $U\Sigma^p V^\top$ maximizes linear descent under a Schatten- $(1 + 1/p)$ constraint [8]; polar Muon is the operator-norm endpoint.

Proposition 6.1 (Exact descent identity). *For every nonzero matrix gradient and $p \in [0, 1]$,*

$$\langle G_B, \Phi_p(G_B) \rangle_F = \sqrt{d_o d_i} \frac{\sum_{j:\sigma_j > 0} \sigma_j^{p+1}}{(\sum_{j:\sigma_j > 0} \sigma_j^{2p})^{1/2}} > 0. \quad (42)$$

Consequently the continuous block update $\dot{B} = -\eta_B \Phi_p(G_B)$, combined with metric descent on nonmatrix blocks, strictly decreases the loss away from a stationary point.

Proof. Unitary invariance of the Frobenius inner product gives $\langle U\Sigma V^\top, U\Sigma^p V^\top \rangle_F = \sum_j \sigma_j^{p+1}$. Substitution of the normalization in (41) proves the identity. Summing $\langle \nabla_B \mathcal{L}, \dot{B} \rangle_F$ over blocks gives strict dissipation. $\square$

The descent identity does not say which Fourier sector advances. That requires an alignment statement between function-space sectors and matrix singular vectors. The next closure is exact when stated and deliberately strong; the experiments measure its failures through signed mode velocity and event coverage.

Assumption 6.2 (Orthogonal singular-sector closure). *On a time interval, one active matrix block has residual coordinates $r_q = \langle f - y, e_q \rangle$ and coefficient Jacobians*

$$J_q := \nabla_B \langle f, e_q \rangle = c_q u_q v_q^\top, \quad u_q^\top u_\ell = v_q^\top v_\ell = \mathbf{1}\{q = \ell\}, \quad (43)$$

with $c_q > 0$. Other blocks and cross-sector forcing are negligible. During the half-recovery interval, the common normalization

$$Z_p(t) = \frac{(\sum_\ell |r_\ell(t) c_\ell|^{2p})^{1/2}}{\eta_B \sqrt{d_o d_i}} \quad (44)$$

is bounded above and below by positive constants independent of q .

Theorem 6.3 (Conditional Muon^p sector clock). *Under Assumption 6.2, the exact residual dynamics are*

$$\dot{r}_q = -\frac{c_q^{p+1}}{Z_p(t)} \text{sign}(r_q) |r_q|^p. \quad (45)$$

For $0 \leq p < 1$, the time to reduce $|r_q|$ from a_q to $a_q/2$ is, up to the two uniform bounds on Z_p ,

$$T_{1/2,q}^{(p)} \asymp c_q^{-(p+1)} a_q^{1-p} \frac{1 - 2^{p-1}}{1 - p}. \quad (46)$$

For $p = 1$, the corresponding clock is $T_{1/2,q}^{(1)} \asymp c_q^{-2} \log 2$. Therefore, if $c_q \asymp q^{-s}$ and $a_q \asymp q^{-\alpha}$,

$$T_{1/2,q}^{(p)} \asymp q^{s(p+1)-\alpha(1-p)}, \quad p < 1, \quad T_{1/2,q}^{(1)} \asymp q^{2s}. \quad (47)$$

Proof. Under (43), $G_B = \sum_q r_q c_q u_q v_q^\top$ is already an SVD up to the signs of r_q . Pairing $-\Phi_p(G_B)$ with J_q gives (45). Integrating $d|r_q|/|r_q|^p$, and bounding $Z_p(t)$ above and below, gives (46). Substitution of the two power laws yields (47). The $p = 1$ case is the logarithmic limit. $\square$

For bounded-variation tangent features, c_q is the Frobenius amplitude of a coefficient Jacobian and c_q^2 is its block curvature. Saturation of the Fourier bound corresponds to $s = 1$; faster decay has $s > 1$. In the saturated ideal closure, gradient flow has a quadratic half-error front, while $p < 1$ reduces the exponent and also gives relatively more motion to small-amplitude target modes. This is the precise sense in which Muon can shorten a Fourier leap. It is not unconditional: different modes need not occupy orthogonal singular frames, the normalization is global, several blocks compete, and a spectral update may initially move a coefficient in the wrong direction through cross-mode forcing. More explicitly, relative to the gradient-flow exponent,

$$2s - [s(p+1) - \alpha(1-p)] = (1-p)(s+\alpha) > 0, \quad p < 1, \quad (48)$$

so every nontrivial spectral flattening advances the asymptotic front under the closure. The strict inequality belongs to the closure, not to arbitrary finite networks.

The $p = 1$ member of (41) is block-normalized gradient descent, not literally the unnormalized maximal-update gradient flow used in the baseline. Under the uniform bound on Z_1 , normalization changes a common clock prefactor but not its q^{2s} exponent. Direct scalar-metric gradient flow gives $\dot{r}_q = -\eta c_q^2 r_q$ under the same sector closure and therefore the same frequency exponent.

For the experimental spectrum $a_q \asymp q^{-1.25}$ and the saturated regularity value $s = 1$, the ideal closure predicts clock exponents $-.25, .50, 1.25$ for $p = 0, 1/3, 2/3$, respectively, compared with 2 for gradient flow. The negative polar exponent is not a claim that harder frequencies are intrinsically easier: a relative-error threshold shrinks with a_q , and the conclusion relies on bounded common normalization and singular-sector alignment. The measured front is therefore compared with this ordering, not forced to equal these ideal numbers.

6.1 What a Muon DMFT must contain

Full-training gradient-flow DMFT closes a two-time activation correlation and backward response at every layer [5]. Replacing a linear matrix-gradient update by (41) changes the response channel itself. For a full-row-rank block, let $S = GG^\top$ and $b = (p-1)/2$, so that the unnormalized map is $\Psi_p(G) = S^b G$. Its exact Fréchet derivative is

$$D\Psi_p(G)[\Delta] = D(S^b)[G\Delta^\top + \Delta G^\top]G + S^b \Delta, \quad (49)$$

where $D(S^b)$ is the Loewner divided-difference operator. Differentiating the normalization additionally projects out the radial component,

$$D\Phi_p(G)[\Delta] = \frac{\sqrt{d_o d_i}}{\|\Psi_p\|_F} \left(D\Psi_p[\Delta] - \Psi_p \frac{\langle \Psi_p, D\Psi_p[\Delta] \rangle_F}{\|\Psi_p\|_F^2} \right). \quad (50)$$

This nonseparable matrix derivative enters the retarded response or Onsager reaction. A reused-data Muon DMFT must therefore close layerwise two-time correlations and responses together with

the evolving singular law and a matrix-valued memory kernel. The three-layer program of Aiad [2] derives this finite response interface and reports earlier Hermite-sector clocks, but explicitly leaves the joint reused-data DMFT closure conditional. Our use is equally limited: the exact finite descent law and [Theorem 6.3](#) are proved, whereas the self-averaging Muon process is a physics closure tested by width transfer. Calling the ordinary spectral update AMP would be incorrect because it contains no self-consistent Onsager subtraction.

7 Experiments

We use the exact architecture in [Section 2](#) on a uniform periodic grid and explicit full-batch gradient descent, with no momentum, adaptive preconditioner, or stochasticity. The primary hierarchy has $(q_j) = (1, 4, 8, 16)$, amplitudes $(1, .65, .45, .30)$, width $m = n = 256$, and rank ratio $\rho = .25$. Rich runs use $\gamma = 1$; lazy controls use $\gamma = 10^{-4}$. The Euler step is 5, well below the initial linear-stability limit, and trajectories run to gradient-flow time 25,000. Unless noted otherwise, uncertainty is the standard deviation over three initializations.

The instantaneous Fourier matrix is computed without forming a sample-space kernel. For each mode we differentiate its output coefficient and evaluate (23). This diagnostic includes the μP metric. The experiments comprise hierarchy, frequency, fixed-gap, rank, width, and spectrum controls. Every trajectory and plotting command is released with the paper.

Figure 1 demonstrates the central separation. The rich model is manifestly non-kernel: its latent representation moves by 3.35 ± 0.08 initial latent norms in the post- $q = 1$ spectrum experiment, and $35.7\% \pm 1.3\%$ of outer gates change state. The lazy values are 3.13×10^{-4} and 9.2×10^{-5} , respectively. Yet the mode-recovery ordering remains low to high.

After fitting the first mode to loss below 3.2×10^{-7} , the rich curvature obeys $\Lambda_{qq} \propto q^{-2.132}$ over $q \geq 3$; the lazy exponent is -2.134 . Thus the exponent is a geometric regularity effect, not evidence of a frozen kernel. The prefactor is dynamic: relative to initialization, rich training multiplies the curvature at $q = 1, 4, 8, 16, 32$ by approximately 1.20, 0.81, 0.69, 0.83, 0.84, while lazy ratios remain near one. The full Fourier matrix has off-diagonal Frobenius ratio 0.136 in the rich regime, confirming that exact diagonalization should not be assumed.

For targets $\cos x + 0.5 \cos(qx)$, rich 20%-recovery times at $q = 2, 4, 6, 8$ are approximately 250, 500, 2750, 9917; modes $q \geq 12$ do not reach the threshold by time 25,000. The lazy baseline reaches $q = 24$ near time 7000. This does not contradict [Theorem 4.2](#): rich dynamics decrease the relevant high-frequency curvatures, increasing the effective plateau constant. Holding the raw gap at three while translating the pair verifies that the difficulty is not a function of the gap alone.

Rank ratios from .0625 to 1 change the mean $q = 8$ recovery time from roughly 11,250 to 9,333, with non-monotone finite-width variation at intermediate ratios. We therefore do not claim that low rank by itself creates sequential learning or that there are exactly r plateaux. Rank changes the DMFT response strength and finite-channel noise; Fourier regularity creates the saddle hierarchy.

The width comparison in [Figure 4](#) supports the extensive-rank DMFT interpretation. It is not, by itself, a proof of convergence, and the remaining deviations are compatible with the $O(m^{-1/2})$ fluctuation theory of Bordelon and Pehlevan [6].

7.1 Full training across depth and architecture

The second campaign uses the models of [Section 2.1](#). Every parameter block is trained. The population grid has 128 points, the main width is 128, MMNN rank ratio is .25, and affine depths are $D \in \{3, 5, 7\}$. The sparse target retains frequencies $(1, 4, 8, 16)$. A second target has the power-law

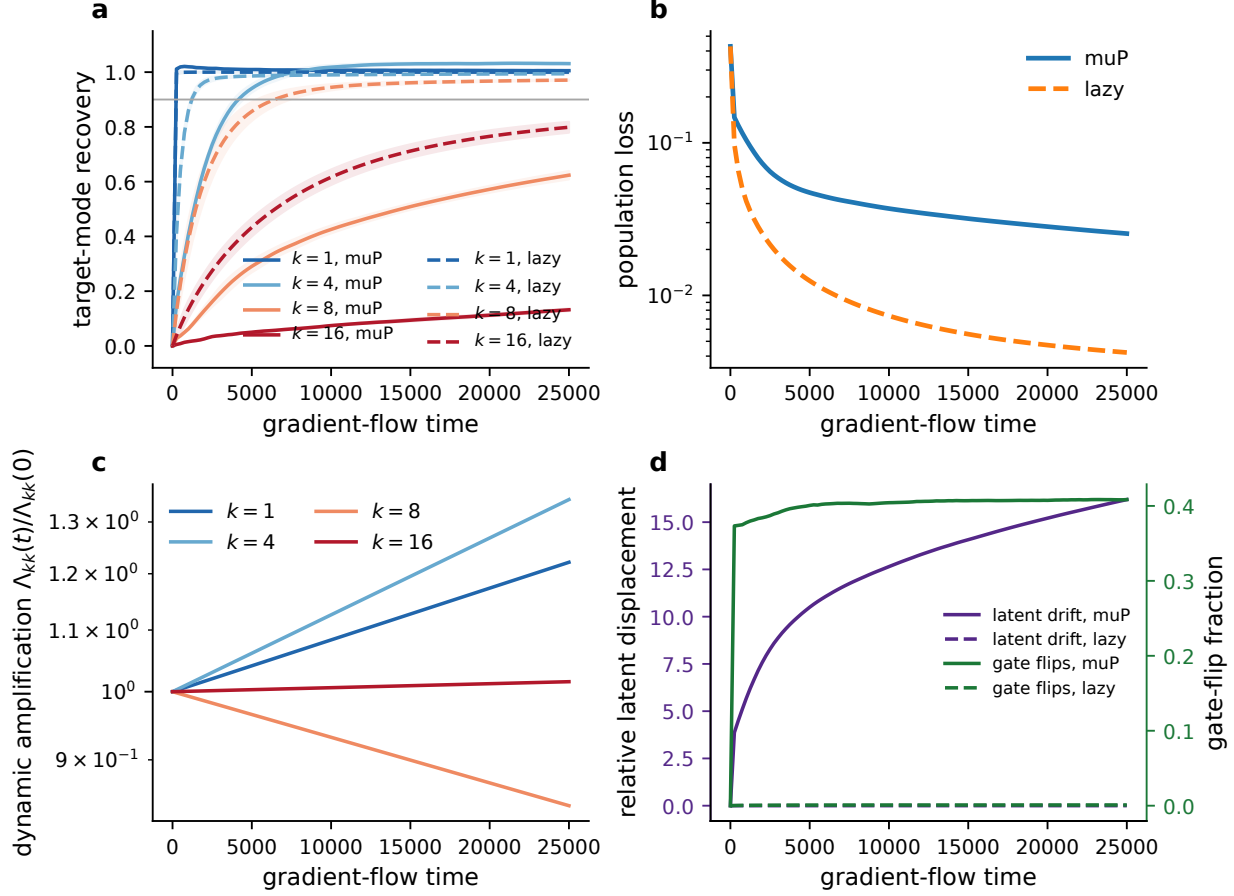


Figure 1: **Hierarchical recovery with order-one feature learning.** (a) Fourier recovery for the sum of four cosines; solid curves are masked μP and dashed curves are lazy. (b) Population loss. The lazy control learns the high modes faster in this masked architecture. (c) Rich-regime tangent-curvature renormalization. The kernel changes non-uniformly across frequencies, so the trajectory is not described by K_0 . (d) Latent displacement (left axis) and gate flips (right axis). Curves show means over three seeds.

spectrum

$$y_{\text{PL}}(x) = \sum_{q=1}^{24} q^{-1.25} \cos(qx). \quad (51)$$

We define the sector event as the first stored step at which $|\hat{f}_q - a_q|/|a_q| \leq 1/2$. Unreached events are right-censored at the common horizon.

Learning rates are selected separately for architecture, depth, and optimizer on seeds 101 and 102, which are absent from confirmation. The prespecified candidate set is $\{.1, .3, 1\}$ for maximal-update gradient descent and $\{.003, .01, .03\}$ for every spectral power. The calibration score averages log relative sector error and uses log endpoint loss only as a light tie breaker. Confirmation uses eight paired seeds. Spectral updates use a direct compact SVD of each matrix gradient (the accurate `gesvd` driver on CUDA), with relative numerical-rank threshold 10^{-7} ; vectors and biases use metric descent. There is no momentum, weight decay, minibatch noise, or Newton–Schulz approximation. Thus this is a mechanism experiment rather than a claim about every implementation marketed

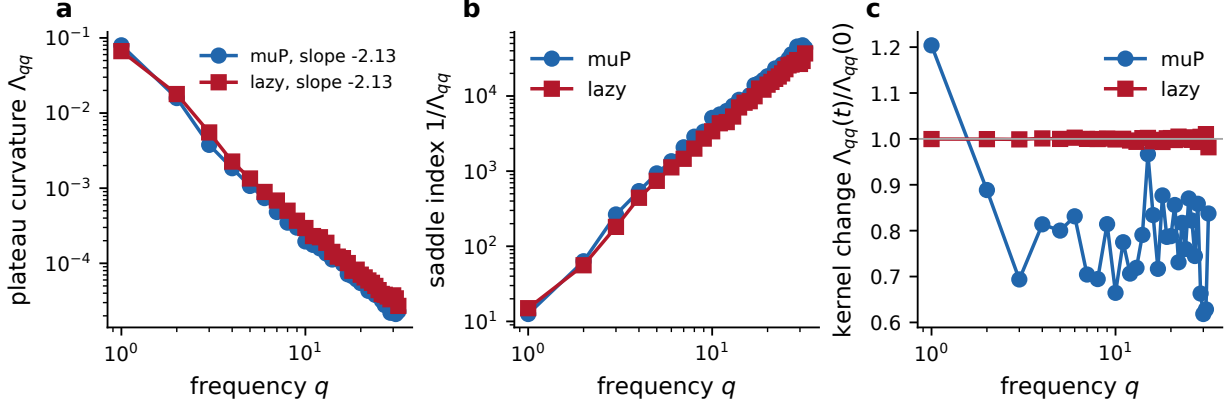


Figure 2: **A hierarchy of dynamic Fourier saddles after fitting $k = 1$.** (a) Instantaneous curvature. Fits over $q \geq 3$ give exponents -2.13 in both regimes. (b) The inverse-curvature saddle index grows approximately quadratically. (c) Rich feature learning changes the kernel by order one and frequency-selectively, whereas the lazy kernel stays at initialization. Means over three seeds are shown.

as Muon. Training and spectral factorization use single precision throughout; this is why the direct-SVD backend audit is part of the acceptance criteria rather than an implementation footnote.

Calibration selected dense gradient-descent rate .3 at all three depths and dense spectral rate .01 for all three powers. For the MMNN it selected gradient-descent rates .3, .3, .1 at depths three, five, and seven and spectral rate .003 for all three powers. Width controls reuse the depth-five rate verbatim. Throughout, “earlier” means fewer calibrated full-batch updates. A direct SVD is substantially more expensive per update than gradient descent; the experiment makes no wall-clock acceleration claim.

Figure 5 rules out the frozen-left mask as the source of the hierarchy. Dense and factorized networks both display successively later sector events while every forward and backward field changes. The claim is deliberately not that deeper is always slower: depth changes the variation budget, cross-mode transfer, and conditioning, so curves may cross.

At depth five the dense median half-error steps are 25, 225, 500, with the $q = 16$ median censored at 2500; the full-training MMNN gives 25, 150, 325, 1500. Median maximum hidden-feature displacement is 5.65 and 8.63 initial-feature norms, respectively. Meanwhile the mean curvature slope over $q \geq 4$ moves from -2.02 to -2.34 in the dense network and from -2.03 to -1.31 in the MMNN, while the $q \geq 4$ curvatures grow by roughly two or more orders of magnitude. Thus feature learning can reshape both slope and prefactor; what persists is a decreasing high-frequency supply, not a frozen or universal fitted exponent. The non-kernel diagnostic is not confined to depth five: median maximum feature displacement across depths three, five, and seven is 3.24, 5.65, 10.91 for the dense network and 5.25, 8.63, 10.02 for the MMNN. At every one of these six architecture–depth combinations, at least one Fourier curvature changes by a factor exceeding $e^{4.8}$.

On the sparse dense target, the gradient-descent clock is (25, 225, 500, > 2500) at $q = (1, 4, 8, 16)$. Polar Muon changes it to (50, 100, 150, 325). Muon^{1/3} gives (25, 50, 175, 700), and Muon^{2/3} gives (25, 75, 275, 1475); all spectral medians are observed.

For the MMNN, the four gradient-descent clocks are 25, 150, 325, and 1500 steps; polar Muon gives 75, 125, 225, and 450. The fractional powers $p = 1/3$ and $p = 2/3$ give (25, 75, 200, 950) and (25, 100, 275, 2000), respectively. The last value has event coverage $7/8$. Thus flattening is not a

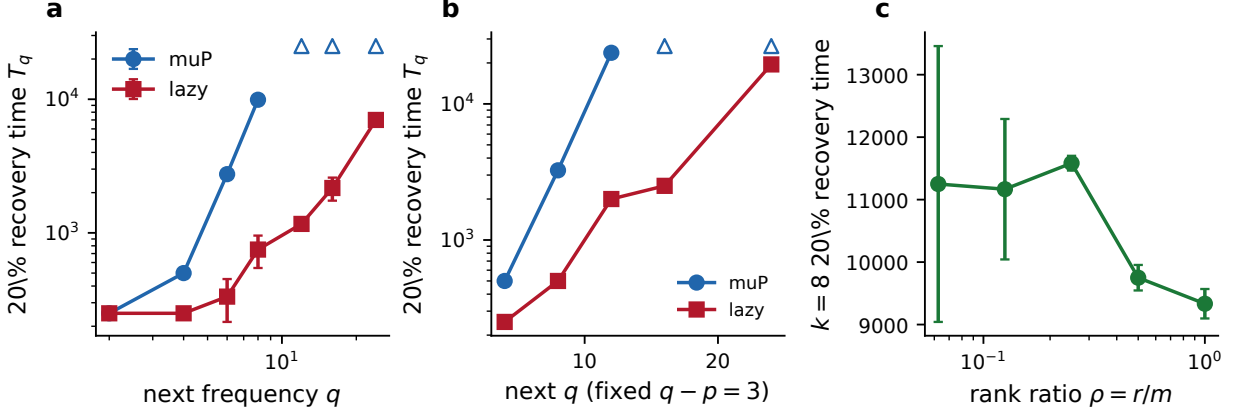


Figure 3: **Frequency, gap, and rank controls.** (a) Time to recover 20% of the next mode when the predecessor is fixed at $p = 1$. Unreached rich-regime modes are right-censored at the horizon and omitted from the connecting line. (b) The raw gap is held exactly at three while both frequencies are translated: difficulty still grows with the absolute next frequency. Open triangles denote censoring. (c) Rank ratio changes the $k = 8$ time scale but not the existence of the hierarchy; error bars show one standard deviation over three seeds.

coordinatewise theorem: it strongly advances the weak sparse sectors for $p = 0$ and $p = 1/3$, while $p = 2/3$ delays the MMNN $q = 16$ median.

The power-law result is more stringent. In the dense network, the $(q = 8, 16, 24)$ gradient-descent clocks (288, 744, 1236) become (132, 252, 504) under polar Muon, with full coverage. $\text{Muon}^{1/3}$ gives (144, 468, > 1500) and $\text{Muon}^{2/3}$ gives (204, 732, > 1500); their $q = 24$ coverages are $3/8$ and $2/8$. In the MMNN, gradient descent gives (168, 432, 564), whereas the three spectral powers give (180, 300, 576), (144, 384, 852), and (252, 684, 1092). The descriptive log-log front slopes are respectively 1.39, .92, 1.68, 1.86 in the dense network and 1.18, .74, 1.31, 1.51 in the MMNN, ordered as gradient descent, $p = 0$, $p = 1/3$, and $p = 2/3$. Polar Muon flattens the measured front exponent, but the ideal exponents of (47) do not quantitatively describe all fractional powers. This is evidence against, rather than an unreported exception to, literal orthogonal-sector closure.

Endpoint risk reverses the sparse clock advantage on the power-law target. For sparse targets, the paired median log-loss differences of polar Muon relative to gradient descent are -2.72 in the dense network and -2.48 in the MMNN, with all $8/8$ pairs favoring Muon. For power-law targets they are $+.55$ (bootstrap 95% interval $[-.03, .59]$, one win) and $+.78$ $[-.49, 2.08]$, three wins), respectively. $\text{Muon}^{1/3}$ and $\text{Muon}^{2/3}$ each lose all sixteen power-law endpoint pairs. The clock effect is therefore real but is neither a fixed-horizon risk guarantee nor an optimizer tournament result.

The sparse and power-law panels support the mechanism of Theorem 6.3: reducing p makes weak singular sectors less subordinate to the leading sector. They do not validate the orthogonal closure literally. In several runs the signed initial velocity of a target coefficient is negative, and the optimizer ranking changes between dense and factorized models. Those are direct signatures of cross-sector and cross-block coupling omitted from Assumption 6.2. This separation between an earlier access clock and endpoint dominance also appears in the twenty-pair three-layer Hermite experiment of Aiad [2]: spectral updates moved median degree-wise events earlier, while every paired endpoint interval included zero and event coverage was equal or lower. The agreement is mechanistic rather than a shared asymptotic theorem. Figure 8 reports the endpoint estimand separately; the paired log-loss differences and bootstrap intervals are given in the machine-readable

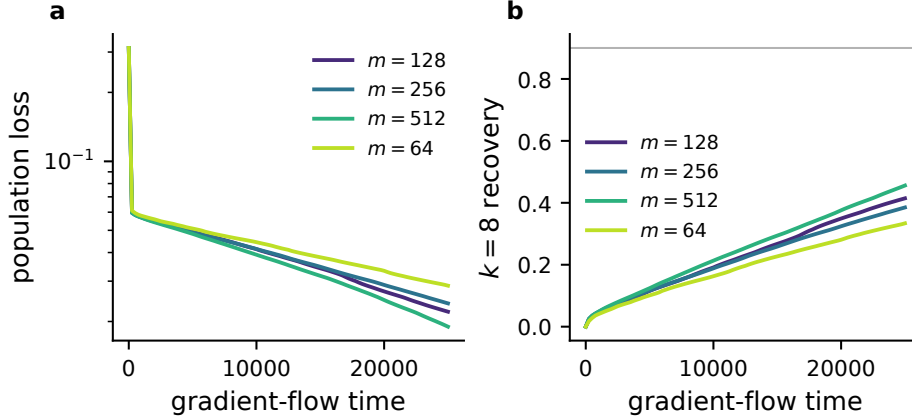


Figure 4: **Width consistency in masked μP .** Population loss and $k = 8$ recovery for $m = n \in \{64, 128, 256, 512\}$, with $r/m = .25$. The curves approach a common rich-regime trajectory rather than shifting their time scale with width.

analysis rather than converted into unpaired error bars.

Figure 11 is a sensitivity result, not a cosmetic overlay. The largest architecture–optimizer median base-to-half change is a factor 4.11 in squared population loss. Because the loss is one half the squared L^2 residual, this is compatible with first-order function error when the continuum endpoint is small. From the quarter to the eighth step, the largest group-median residual-norm ratio is 1.61; doubling the population grid changes it by at most 1.33. Every base, half, quarter, eighth, and fine-grid control retains the ordered $q = 4, 8, 16$ clock. Dense Muon^{1/3} advances at least one of $q = 8, 16, 24$ in every control; at the eighth step its ($q = 8, 16, 24$) clocks are (320, 560, 1200) updates, versus (1280, 3360, 7120) for gradient descent. The MMNN result is less uniform: the half-step run has no strict weak-sector advantage, although the eighth-step clocks (480, 800, 1360) beat gradient descent (400, 1200, 2320) at $q = 16, 24$ but not at $q = 8$. Thus numerical refinement supports the hierarchy and the dense speedup, while confirming that the small MMNN power-law advantage is discretization-sensitive.

The top row of Figure 9 is particularly important for the NTK objection. The tangent curvatures change by order one or more under full training, so the observed clock cannot be generated by freezing the initial kernel. The fitted frequency slope can itself change substantially, as it does in the MMNN. The bounded-variation theorem is an instantaneous upper envelope with an evolving budget, not a claim that every finite-range fit has exponent exactly -2 . The optimizer changes the trajectory through the nonlinear block map, while regularity continues to constrain the instantaneous supply of high-frequency tangent features.

8 Adversarial audit of the claims

We record the strongest objections and the corresponding limits of the result.

“This is still just an NTK proof.” No. The identity $\dot{f} = -\mathcal{K}_t(f - y)$ holds for any differentiable parameterization. An NTK proof replaces K_t by K_0 . We never do so in Theorem 4.2; $\mathcal{B}_t$ and $\Lambda(t)$ may change at order one. Empirically, latent motion, gate flips, and frequency-dependent kernel changes directly reject the frozen-kernel approximation.

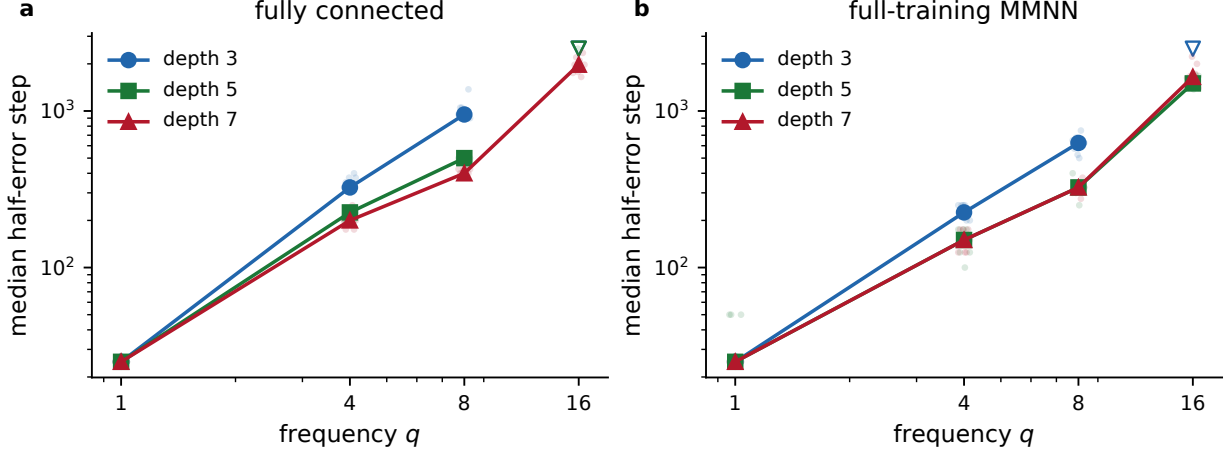


Figure 5: **The Fourier hierarchy survives full training and depth.** Median half-error steps under maximal-update gradient descent for affine depths three, five, and seven. The left panel is fully connected; the right panel trains both factors of every signed low-rank MMNN map. Open downward triangles denote a median that is right-censored at the training horizon. The ordered clock is present across architectures, while depth changes its prefactor and event coverage.

“Calling a plateau a saddle is imprecise.” Agreed if “saddle” means an exact stationary point with a negative Hessian direction. Our definition (28) is an approximate-saddle notion: nonzero residual with a parametrically small gradient. The theorem proves this notion and no stronger Morse-theoretic claim.

“The residence-time theorem assumes the conclusion.” It assumes a quantitative leakage inequality, not sequential learning. The inequality is necessary because task adaptation destroys translation invariance and permits Fourier transfer. Without it, only the instantaneous gradient theorem is unconditional. The measured off-diagonal ratio is small but nonzero; a fully uniform verification of (29) over arbitrarily long time remains open.

“A frequency gap should be $q - p$.” The fixed-gap control and Corollary 4.8 show why this is not an invariant difficulty measure. The Fourier integration-by-parts bound sees q , not a label relative to a previously fitted mode.

“This is precisely the leap framework of Abbe et al. [1].” No. Their theorem concerns online SGD, high ambient dimension, and new coordinate support in a tensor-product Hermite/Fourier–Walsh basis. Our population circle problem has none of those scaling variables. More decisively, Proposition 4.9 shows that their definition assigns the one-coordinate Hermite analogue the minimum degree, independently of all later frequencies. What transfers is the sequential-learning metaphor, not the complexity theorem.

“The result is caused by Adam.” No experiment uses Adam. The geometric campaign uses deterministic full-batch gradient descent. The optimizer campaign applies a declared matrix spectral map, so its difference from gradient descent is the object being tested rather than an unreported adaptive preconditioner.

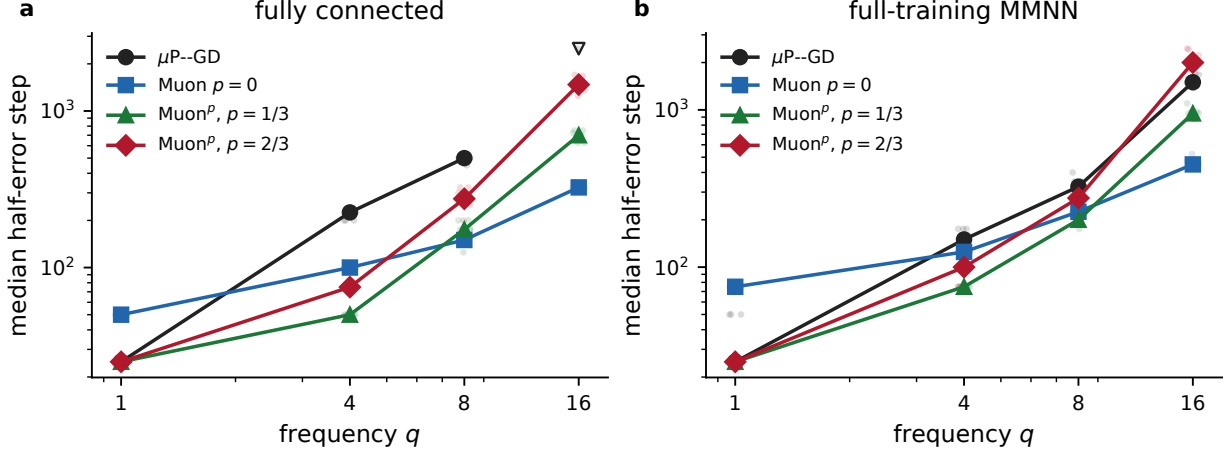


Figure 6: **Spectral powers move sparse-target sector clocks.** Median half-error steps at affine depth five for maximal-update gradient descent, polar Muon, and Muon^p with $p = 1/3, 2/3$. Optimizers are calibrated on disjoint seeds and compared on paired initializations. Open downward triangles mark a right-censored median at the common horizon. Earlier crossings do not imply lower terminal risk or complete event coverage.

“You only confirmed a frozen-left three-map network.” That objection applied to the first campaign and motivated the second. The full-training study updates every block in dense and factorized networks at affine depths three, five, and seven. It confirms the ordering empirically; [Corollary 4.3](#) supplies the finite-depth theorem. Neither result is a uniform-in-depth limit, and the variation-budget constant may worsen with depth.

“Muon is guaranteed to erase the saddles.” False. [Proposition 6.1](#) guarantees total first-order descent, not positive velocity in every Fourier coefficient. The clock exponent in [Theorem 6.3](#) requires biorthogonal singular sectors and controlled normalization. Negative measured coefficient velocities and incomplete event coverage falsify a universal per-mode acceleration claim.

“This is production Muon or an optimizer tournament.” It is neither. We use direct memoryless SVD powers and omit momentum, Newton–Schulz error, weight decay, and wall-clock matching. Candidate learning rates are calibrated on disjoint seeds under a shared protocol, which makes sector clocks interpretable but does not decide which optimizer is best under a compute-matched hyperparameter budget.

“Bordelon–Pehlevan DMFT already proves the Muon result.” Their self-consistent DMFT proves rich dynamics for the declared gradient-based computation graph. A spectral matrix channel is nonseparable and its response is (50). Reused-data training additionally generates retarded memory. The joint Muon closure is conditional here and in Aiad [2]; width transfer is only an empirical check.

“Forming $G^\top G$ is numerically equivalent to a direct SVD.” Not for the polar member in single precision. The Gram construction squares the condition number and, in our audit, changed the polar direction substantially in small singular sectors. Across nine representative square gradient blocks, its median cosine with the direct polar direction was .537 and its median relative error

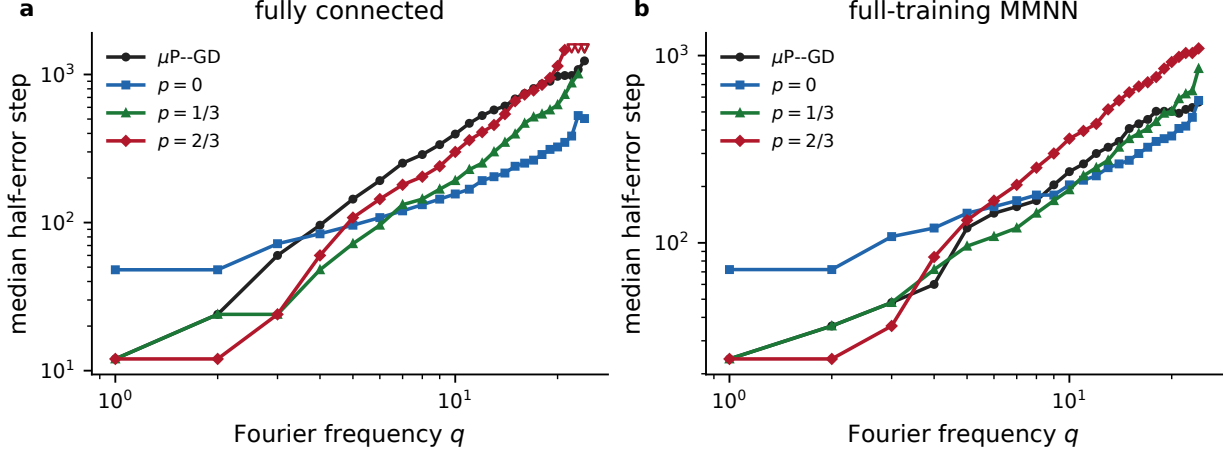


Figure 7: **Muon reshapes a power-law Fourier front.** Median half-error step for every component of (51). Polar flattening reduces the descriptive front slope in both architectures, but absolute clocks, event coverage, and the best power are architecture-dependent. Fractional powers can advance intermediate sectors while delaying or censoring the last ones. Open downward triangles mark right-censored medians; no high-frequency event is imputed.

was .963. We therefore exclude those pilot outputs, use a direct SVD in confirmation, record its numerical-rank threshold, and regression-test an ill-conditioned matrix. This numerical issue matters less as p approaches one but cannot be ignored at $p = 0$.

“Low rank means fixed rank.” Not in the deterministic DMFT claim. We use an extensive bottleneck $r/m \rightarrow \rho < 1$. Fixed rank is explicitly excluded from Proposition 3.2, though Theorem 3.1 and Theorem 4.2 remain valid at finite rank.

“The experiment proves the DMFT equations.” It does not. Width collapse and concentration of measured order parameters validate predictions of the established wide-limit framework in this architecture. A direct numerical solution of the full two-time response equations would be a stronger comparison.

“HCIZ should produce the saddle hierarchy.” HCIZ controls a static rotational integral in a proportional Bayesian teacher–student problem. It does not provide the training-time kernel G_t of our quenched, masked gradient flow. Importing it would add machinery without closing the required dynamics.

9 Discussion

The source of the observed frequency leaps is a product of geometry, dynamics, and optimizer geometry. Bounded-variation tangent features have Fourier coefficients of order at most $1/q$, so scalar-metric curvature is at most order $1/q^2$. In the lazy regime this scarcity persists because the kernel is fixed. In masked μP , DMFT evolves the back-propagated gate kernel; in full training, every forward and backward two-time field evolves. A gradient-flow plateau ends by diagonal amplification or off-diagonal transfer. A Muon step adds a third route: it flattens the singular spectrum of the

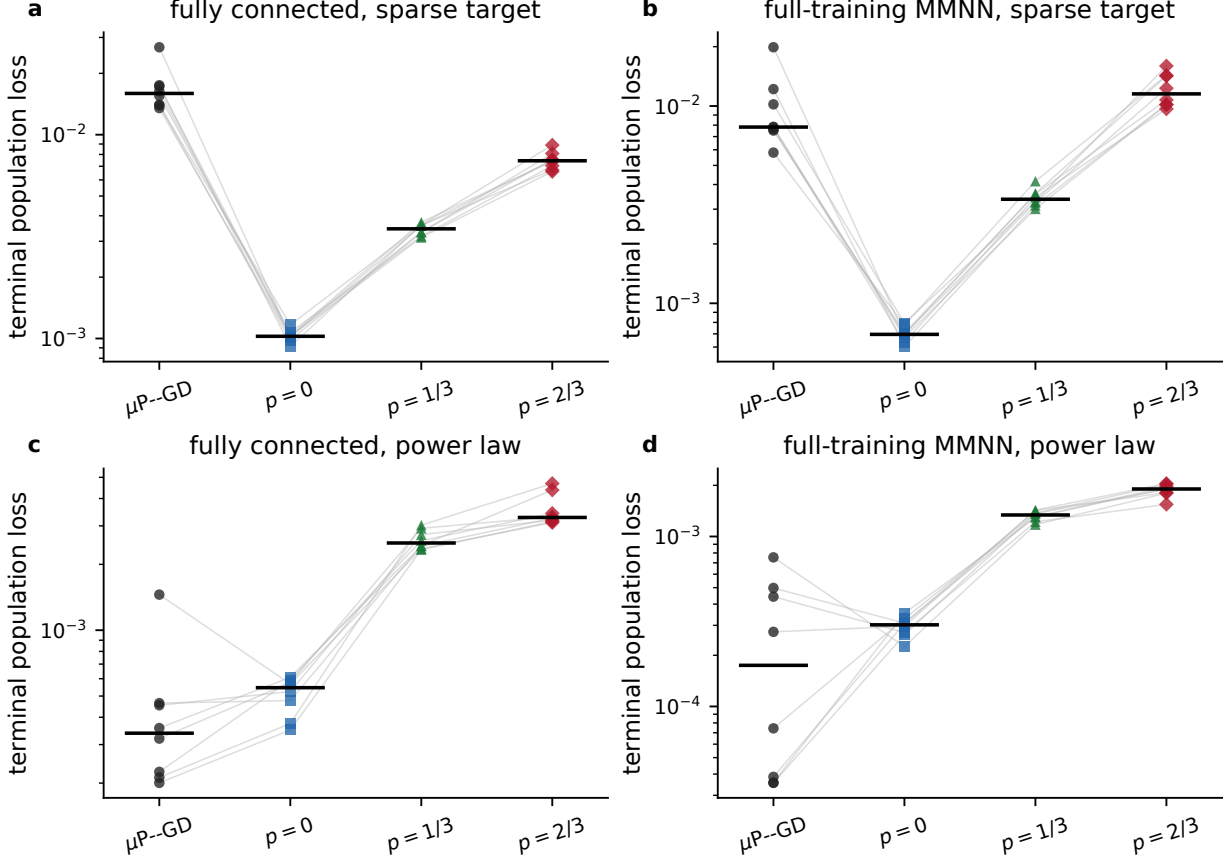


Figure 8: **Earlier access is distinct from endpoint dominance.** Terminal population losses for the same eight initializations under all four update rules. Gray segments connect paired seeds; colored symbols are individual outcomes and black bars are marginal medians. Rows separate the sparse and power-law targets, and columns separate dense and factorized networks. Crossings are retained: a shifted sector clock is not a universal terminal-risk ordering.

complete matrix gradient, giving weak aligned sectors more update mass before the tangent kernel has itself adapted.

This distinguishes four statements that are often conflated:

1. A tangent-kernel identity is exact for all gradient flows.
2. A frozen tangent kernel is an NTK approximation.
3. A deterministic, evolving tangent kernel is a DMFT order parameter in the rich infinite-width limit.
4. A nonlinear spectral update is not described by that kernel alone; its DMFT also needs the Fréchet response of the matrix channel and retarded memory.

Our regularity theorem uses the first, rejects the second, and interprets the third. The Muon analysis addresses the fourth under an explicit singular sector closure. The resulting hierarchy is robust to feature learning, but its residence times are not universal because the tangent fields, singular frames, and cross-mode forcing are task- and architecture-dependent.

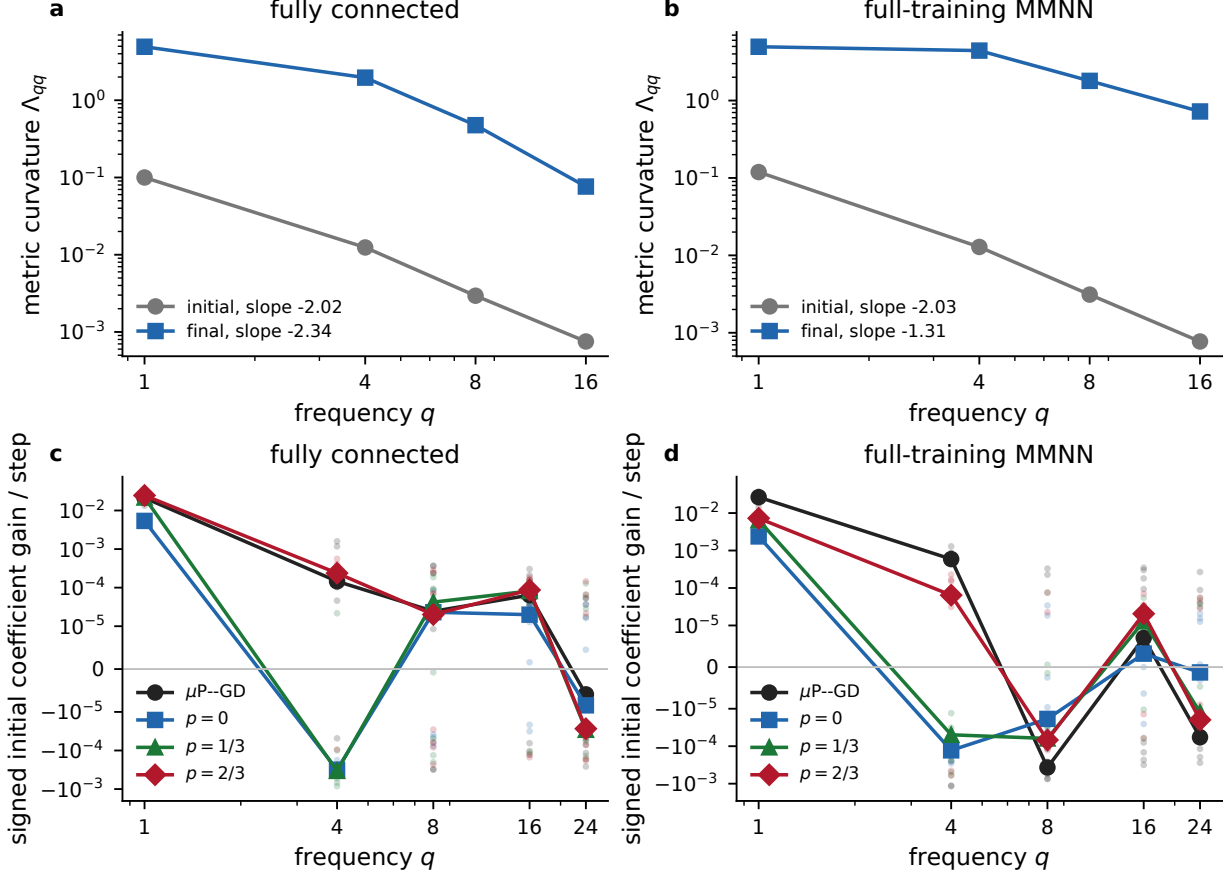


Figure 9: **Dynamic curvature and the failure of a purely diagonal Muon picture.** Top: maximal-update Fourier curvature at initialization and the end of full training. Its prefactor changes by orders of magnitude while the high-frequency supply remains decreasing. Bottom: signed initial coefficient gain per optimizer step on the power-law target; faint points are paired seeds and lines are medians. Negative values expose cross-mode forcing and are retained rather than clipped.

The full-training experiments resolve the first obvious extension: freeing the readout and both MMNN factors changes constants but does not remove the hierarchy. The next theoretical target is a direct numerical solution of the deep two-time DMFT, followed by the nonseparable Muon response closure. Smooth activations replace the bounded-variation exponent by a higher Sobolev exponent, yielding steeper scarcity unless feature learning creates finer scales. Literal nonnegative factorization requires projected dynamics and active-set order parameters. Finally, connecting Fourier plateaux to the high-dimensional coordinate leaps of Abbe et al. [1] requires a joint target with both support discovery and within-support oscillation; neither theory alone implies the other.

10 Conclusion

Frequency learning in the feature-learning regime is not “just an NTK proof,” and it is not an instance of coordinate leap complexity. Across masked and fully trained dense and factorized networks, the kernel and features move at order one while instantaneous tangent regularity continues to make high Fourier modes scarce. This produces a hierarchy of dynamic approximate saddles.

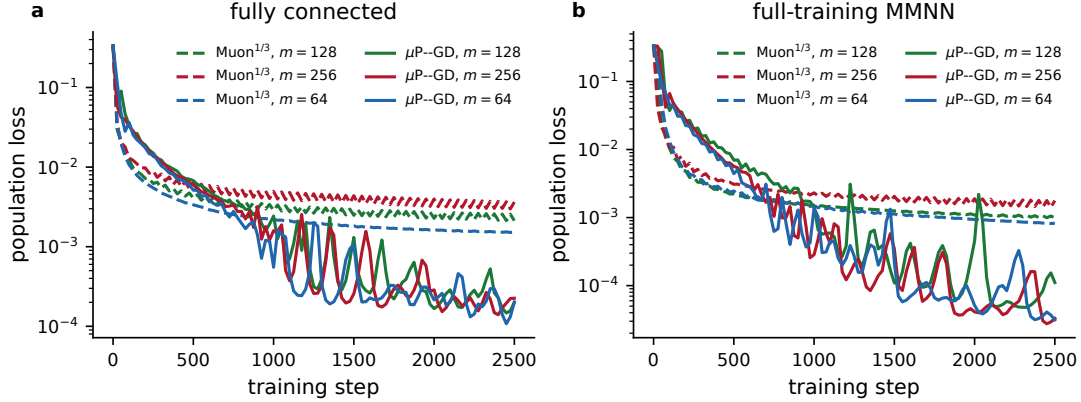


Figure 10: **Width transfer without retuning.** Population loss for widths 64, 128, 256 using the learning rate selected at width 128. Solid curves are maximal-update gradient descent and dashed curves are $\text{Muon}^{1/3}$. The comparison is evidence for a common feature-learning scale, not a proof of a Muon DMFT.

Gradient-flow DMFT governs whether learned fields build the missing tangent features or suppress them. Spectral-power descent can advance a weak Fourier front by reducing the power with which small singular sectors are penalized, but cross-sector forcing prevents a universal per-mode or endpoint guarantee. The correct conclusion is therefore mechanistic rather than promotional: feature learning controls the supply of Fourier directions; Muon controls how the currently supplied matrix-gradient directions share an update.

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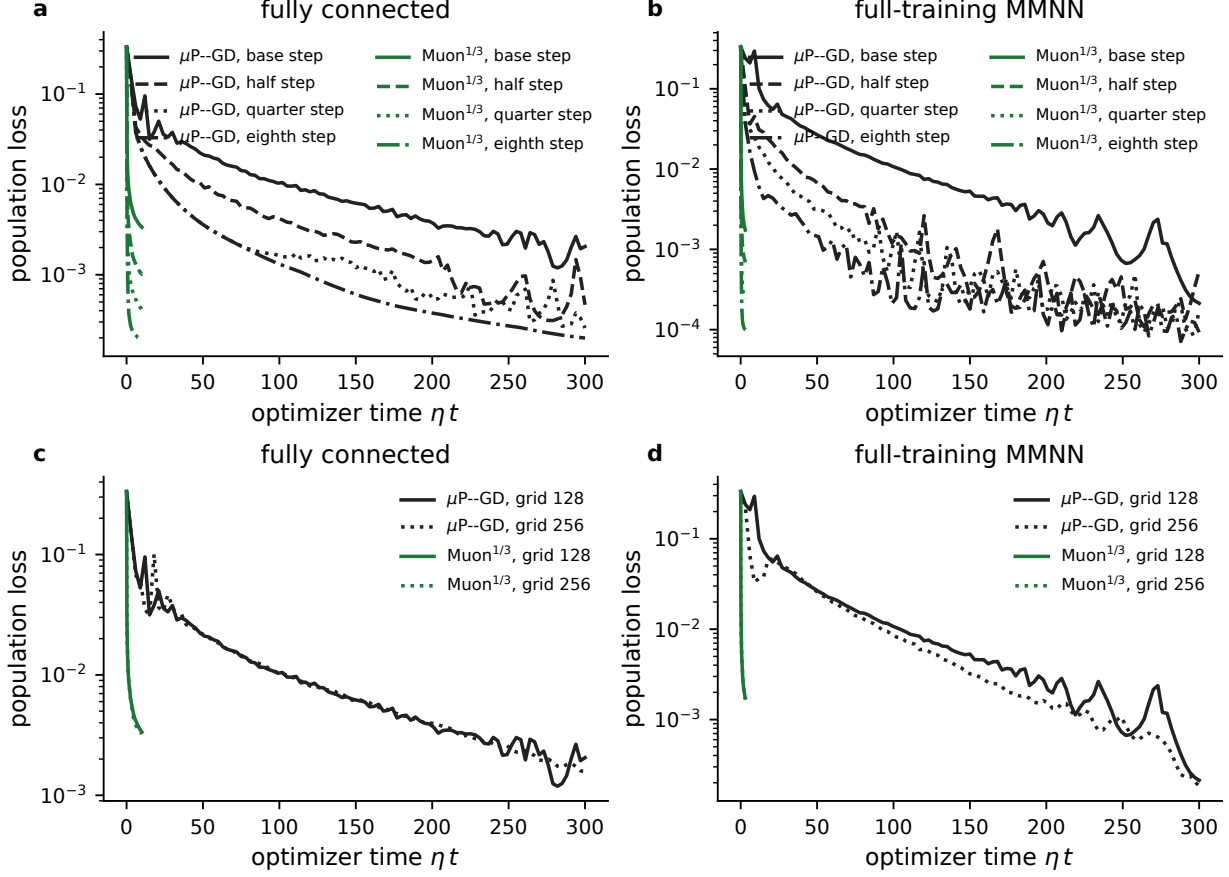


Figure 11: **Step-size and population-grid controls.** Top: the calibrated, half, quarter, and eighth steps, with the horizon increased to keep optimizer time ηt fixed. Bottom: the 128-point population grid is compared with 256 points at the same update rule. Curves are medians over three fresh seeds for maximal-update gradient descent and $Muon^{1/3}$; no rate is retuned for a control.

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A Further derivations

A.1 Fourier coefficient conventions

For $u \in L^2(\mathbb{T})$, write

$$u = u_0 + \sum_{q \geq 1} (u_q^c e_q^c + u_q^s e_q^s), \quad e_q^c = \sqrt{2} \cos(qx), \quad e_q^s = \sqrt{2} \sin(qx).$$

The experiment reports the conventional cosine amplitude $\hat{u}_q^c = 2 \langle u, \cos(q \cdot) \rangle = \sqrt{2} u_q^c$. The tangent curvature in the code includes the factor $1/2$ needed to convert its gradient to the normalized coefficient u_q^c .

A.2 A leakage-free invariant subspace

If $K_t(x, x') = \kappa_t(x - x')$ for all t , Fourier modes diagonalize the kernel and $\varepsilon = 0$ in [Assumption 4.5](#). In that special case

$$R_q(t) = R_q(0) \exp \left(- \int_0^t \Lambda_{qq}(s) ds \right).$$

This formula is exact and non-kernel when $\Lambda_{qq}(s)$ evolves. We do not use it in the main theorem because a fixed cosine target generally breaks translation invariance of the learned kernel.

A.3 Approximate purity

Suppose $R = A e_q + \delta$. Then

$$\|\nabla \mathcal{L}\|_{\gamma^2 n I} \leq |A| \sqrt{\Lambda_{qq}} + \sqrt{\langle \delta, \mathcal{K}_t \delta \rangle}.$$

Thus [Theorem 4.2](#) is stable to a residual perturbation whose kernel energy is small relative to $A^2 \Lambda_{qq}$. An L^2 -small perturbation alone is insufficient if it lies in a very high-curvature direction.

B Reproducibility

The executable entry point is

```
uv run python experiments/mup_dmft_frequency/run_study.py
```

It writes raw training traces, run summaries, instantaneous Fourier matrices, metadata, and all figures. The plotting stage can be repeated without training using `--plots-only`. A short end-to-end smoke test is available through `--quick`. The implementation automatically uses a CUDA device when available and otherwise runs on CPU. Random seeds, parameterization constants, widths, ranks, time steps, and target amplitudes are recorded in the output files.

The full-training and spectral-power campaign is reproduced by

```
uv run python experiments/mup_dmft_frequency/run_full_training_muon.py \
  --mode calibrate
uv run python experiments/mup_dmft_frequency/run_full_training_muon.py \
  --mode campaign
uv run python experiments/mup_dmft_frequency/run_full_training_muon.py \
  --mode discretization
uv run python experiments/mup_dmft_frequency/run_full_training_muon.py \
  --mode discretization-quarter
uv run python experiments/mup_dmft_frequency/run_full_training_muon.py \
  --mode discretization-eighth
uv run python experiments/mup_dmft_frequency/audit_spectral_backend.py
uv run python experiments/mup_dmft_frequency/sync_full_training_figures.py
# rebuild main.pdf here using the commands above
uv run python experiments/mup_dmft_frequency/audit_full_training_results.py \
  --require-discretization
```

Calibration and confirmation use disjoint seeds. The second command writes all paired trajectories, event summaries, width controls, a JSON claim analysis, and the additional figures. The third command halves the optimizer step at matched optimizer time and doubles the population-grid resolution; the fourth and fifth append quarter- and eighth-step refinements after inspecting the declared base-to-half and half-to-quarter sensitivities. The sixth reproduces the rejected Gram-backend comparison, the seventh synchronizes vector figures into the paper tree, and the final command fails closed on the predeclared numerical gates after rebuilding the PDF. A smoke campaign is available by adding `--quick` to both training stages; `--mode plots-only` regenerates the figures and analysis without training.